POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 3042 | Semester III | Theory | 4 Credits

Digital Circuits and Systems

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 3042 as **Digital Circuits and Systems** in Semester III for Electronics Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public digital-electronics curricula and is not presented as recovered official SITTTR Course 3042 detailed-syllabus wording.

Source treatment: The previous generated Course 3042 handbook was corrected because the attached course index confirms this course title as Digital Circuits and Systems. Teaching scheme, credits and assessment values remain provisional placeholders; the modules follow the cited public digital-logic curricula and must be reconciled with restored official SITTTR content.

Verified source note: The official SITTTR detailed source was inaccessible. Public sources supporting Boolean logic, combinational/sequential design, logic families and ADC/DAC are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Digital Circuits and Systems develops binary reasoning and practical logic design. It moves from number systems and Boolean algebra to combinational blocks, sequential circuits, logic families and conversion between analog and digital representations.

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Prerequisite foundation

Basic electronics, binary arithmetic and safe laboratory practice with low-voltage digital ICs.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Apply number systems, Boolean algebra and logic gates to form logic functions.
2. Design and analyse combinational circuits using minimisation and standard building blocks.
3. Explain flip-flops, registers, counters and basic state-machine behaviour.
4. Compare logic families and explain fundamental ADC/DAC operation and selection.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Apply number systems, Boolean algebra and logic gates to represent and simplify logic functions.	Apply
CO2	Analyse and design basic combinational logic blocks using truth tables and minimisation.	Apply
CO3	Explain and design elementary sequential circuits using flip-flops, registers and counters.	Apply
CO4	Explain logic-family characteristics and select basic ADC/DAC methods for an application.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Binary Systems, Boolean Algebra and Gates	Binary/octal/hexadecimal representation, signed numbers, Boolean laws, truth tables, standard forms and logic gates.	Approx. 14 hours	CO1	Module 1
2	Combinational Circuit Design	SOP/POS, K-maps, adders/subtractors, comparators, encoders/decoders, multiplexers/demultiplexers, ROM/PLA/PAL.	Approx. 16 hours	CO2	Module 2
3	Sequential Logic and State Systems	Latches, SR/JK/D/T flip-flops, triggering, excitation tables, registers, shift registers, ripple/synchronous counters and state machines.	Approx. 16 hours	CO3	Module 3
4	Logic Families and Data Conversion	TTL/CMOS/ECL characteristics, fan-in/fan-out, noise margin, propagation delay; DAC/ADC types, specifications and selection.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Binary Systems, Boolean Algebra and Gates

Binary/octal/hexadecimal representation, signed numbers, Boolean laws, truth tables, standard forms and logic gates.

CO1 · Approx. 14 hours

Combinational Circuit Design

SOP/POS, K-maps, adders/subtractors, comparators, encoders/decoders, multiplexers/demultiplexers, ROM/PLA/PAL.

CO2 · Approx. 16 hours

Sequential Logic and State Systems

Latches, SR/JK/D/T flip-flops, triggering, excitation tables, registers, shift registers, ripple/synchronous counters and state machines.

CO3 · Approx. 16 hours

Logic Families and Data Conversion

TTL/CMOS/ECL characteristics, fan-in/fan-out, noise margin, propagation delay; DAC/ADC types, specifications and selection.

CO4 · Approx. 14 hours

MODULE 1

Binary Systems, Boolean Algebra and Gates

Binary/octal/hexadecimal representation, signed numbers, Boolean laws, truth tables, standard forms and logic gates.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

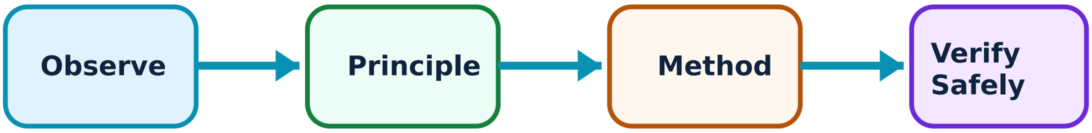
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Binary Systems, Boolean Algebra and Gates: identify the system, state its principle, follow the correct method and verify the result safely.

1. Number systems and codes−

Digital systems represent information with discrete symbols, commonly binary bits. Conversion among binary, octal and hexadecimal and signed representation are essential for interpreting logic-system data.

Study and application method

Write positional weights, show each conversion step and verify by converting the result back or checking decimal value.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write positional weights, show each conversion step and verify by converting the result back or checking decimal value.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: State bit width before interpreting signed values; the same bit pattern can represent different values under different conventions.

2. Boolean algebra and truth tables−

Boolean algebra models two-valued logic. Truth tables specify output for every input combination; algebraic laws and De Morgan’s theorem transform or simplify equivalent expressions.

Study and application method

Define variables and logic level meaning, create the full truth table, then apply one algebraic rule per line while preserving equivalence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Do not simplify by intuition alone; check the final expression against the required truth table.

3. Logic gates and universal implementation−

AND, OR, NOT, NAND, NOR, XOR and XNOR gates implement Boolean functions. NAND and NOR are universal gates because any logic function can be formed from either family.

Study and application method

Start from the required equation/truth table, draw gate-level blocks, label intermediate signals and validate with test vectors.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Gate symbols, active-low bubbles and IC pinouts must be checked carefully before hardware wiring.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Combinational Circuit Design

SOP/POS, K-maps, adders/subtractors, comparators, encoders/decoders, multiplexers/demultiplexers, ROM/PLA/PAL.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

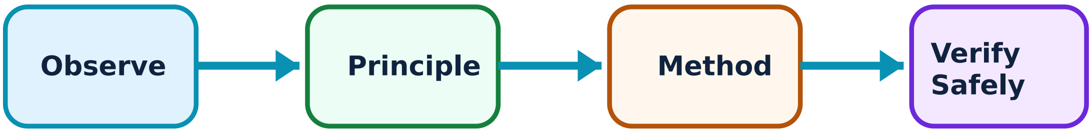
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Combinational Circuit Design: identify the system, state its principle, follow the correct method and verify the result safely.

1. K-map minimisation−

Karnaugh maps provide a visual method to simplify Boolean functions with a modest number of variables. Grouping adjacent 1s or 0s yields reduced SOP or POS expressions.

Study and application method

Map minterms/maxterms using Gray-code order, make largest valid power-of-two groups including edge adjacency, derive each term and verify coverage.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map minterms/maxterms using Gray-code order, make largest valid power-of-two groups including edge adjacency, derive each term and verify coverage.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not group diagonal cells or groups of arbitrary size; only adjacency rules and powers of two apply.

2. Arithmetic and comparison blocks−

Half/full adders, subtractors and comparators combine logic functions to perform binary arithmetic and magnitude comparison. Multi-bit blocks cascade or use dedicated IC structures.

Study and application method

Begin with a one-bit truth table, derive sum/difference/carry/borrow outputs, then explain how the blocks extend to multiple bits.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Begin with a one-bit truth table, derive sum/difference/carry/borrow outputs, then explain how the blocks extend to multiple bits.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Carry/borrow propagation affects multi-bit behaviour; test extreme and carry-in cases.

3. MSI, multiplexing and programmable logic−

Multiplexers select one input; decoders/encoders translate codes; ROM, PLA and PAL implement predefined or programmable logic. These blocks reduce discrete-gate complexity.

Study and application method

Identify data inputs, select inputs, enable and outputs; create a selection table before wiring or programming.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Enable polarity and unused inputs must be tied according to the device datasheet, not assumed.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Sequential Logic and State Systems

Latches, SR/JK/D/T flip-flops, triggering, excitation tables, registers, shift registers, ripple/synchronous counters and state machines.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

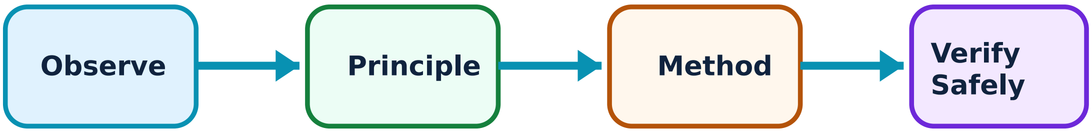
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Sequential Logic and State Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Latches and flip-flops−

Sequential circuits store state. Latches respond to enable level; flip-flops sample on a clock edge. SR, JK, D and T flip-flops differ in allowable input behaviour and use.

Study and application method

Use a truth/characteristic table, label clock/enable edge, trace present state to next state and note setup/hold requirements conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a truth/characteristic table, label clock/enable edge, trace present state to next state and note setup/hold requirements conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Asynchronous input changes near a clock edge can cause unreliable or metastable behaviour.

2. Registers and counters−

Registers store multi-bit words; shift registers move data between stages. Counters progress through a defined state sequence and may be ripple (asynchronous) or synchronous.

Study and application method

Draw state sequence and timing diagram, choose flip-flop type/count, identify reset and unused-state handling, then test the complete cycle.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw state sequence and timing diagram, choose flip-flop type/count, identify reset and unused-state handling, then test the complete cycle.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Ripple counter propagation delays can create transient states that are unsuitable for some decoding applications.

3. State-machine design−

A state machine combines memory and combinational logic so outputs depend on state and possibly inputs. State diagrams/tables make behaviour explicit before implementation.

Study and application method

List states, inputs, outputs and reset state; create transition/output table, choose encoding and verify every input/state combination.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List states, inputs, outputs and reset state; create transition/output table, choose encoding and verify every input/state combination.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not omit invalid or reset/recovery conditions in a practical state-machine design.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Logic Families and Data Conversion

TTL/CMOS/ECL characteristics, fan-in/fan-out, noise margin, propagation delay; DAC/ADC types, specifications and selection.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

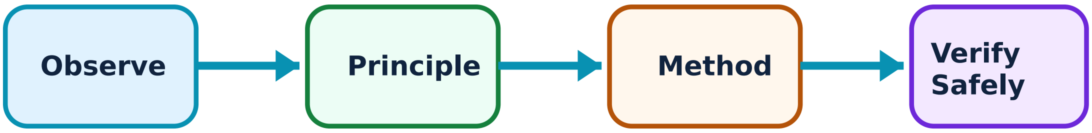
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Logic Families and Data Conversion: identify the system, state its principle, follow the correct method and verify the result safely.

1. Logic-family characteristics−

Logic families differ in supply range, logic thresholds, speed, power, fan-out, noise margin and input/output structure. Device selection balances system requirements.

Study and application method

Compare candidate families using a specification table with voltage, delay, power, interfacing and environmental constraints.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare candidate families using a specification table with voltage, delay, power, interfacing and environmental constraints.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never mix logic families without checking threshold compatibility, supply requirements and output-drive capability.

2. Digital-to-analog conversion−

A DAC converts a digital code into an analog voltage/current. Weighted-resistor, R–2R and PWM approaches have different resolution, accuracy, speed and filtering needs.

Study and application method

State required range, resolution and update speed; calculate code step conceptually and choose a topology with an output-conditioning plan.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State required range, resolution and update speed; calculate code step conceptually and choose a topology with an output-conditioning plan.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Resolution is not the same as accuracy; reference stability and nonlinearity also matter.

3. Analog-to-digital conversion−

An ADC samples an analog signal and converts it to a code. Flash, counter, SAR, sigma-delta and integrating types trade speed, resolution, noise performance and complexity.

Study and application method

Define input range, signal bandwidth, required resolution and conversion rate; select a type and state anti-aliasing/reference considerations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define input range, signal bandwidth, required resolution and conversion rate; select a type and state anti-aliasing/reference considerations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: [illegible] concept [illegible] [illegible] [illegible]. [illegible] diagram / circuit / formula / procedure [illegible] [illegible]. [illegible] result [illegible] application [illegible] [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Common mistake / precaution: Sampling without appropriate signal conditioning can create aliasing or invalid readings.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Binary Systems, Boolean Algebra and Gates

Use the concept flow to revise observation, principle, method and verification.

Module 2: Combinational Circuit Design

Use the concept flow to revise observation, principle, method and verification.

Module 3: Sequential Logic and State Systems

Use the concept flow to revise observation, principle, method and verification.

Module 4: Logic Families and Data Conversion

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Convert and verify binary/octal/hexadecimal values, including a fixed-width signed representation exercise.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Minimise a truth-table function with K-maps and implement it using universal gates in simulation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Build or simulate a flip-flop/register/counter and compare timing with its state table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare a TTL/CMOS choice and select an ADC/DAC topology for a stated sensor or actuator need.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Binary Systems, Boolean Algebra and Gates

Explain Number systems and codes and state one correct method, application or precaution.−

Digital systems represent information with discrete symbols, commonly binary bits. Conversion among binary, octal and hexadecimal and signed representation are essential for interpreting logic-system data.

Answer framework: Write positional weights, show each conversion step and verify by converting the result back or checking decimal value.

Important point: State bit width before interpreting signed values; the same bit pattern can represent different values under different conventions.

Explain Boolean algebra and truth tables and state one correct method, application or precaution.−

Boolean algebra models two-valued logic. Truth tables specify output for every input combination; algebraic laws and De Morgan’s theorem transform or simplify equivalent expressions.

Answer framework: Define variables and logic level meaning, create the full truth table, then apply one algebraic rule per line while preserving equivalence.

Important point: Do not simplify by intuition alone; check the final expression against the required truth table.

Explain Logic gates and universal implementation and state one correct method, application or precaution.−

AND, OR, NOT, NAND, NOR, XOR and XNOR gates implement Boolean functions. NAND and NOR are universal gates because any logic function can be formed from either family.

Answer framework: Start from the required equation/truth table, draw gate-level blocks, label intermediate signals and validate with test vectors.

Important point: Gate symbols, active-low bubbles and IC pinouts must be checked carefully before hardware wiring.

Module 2 — Combinational Circuit Design

Explain K-map minimisation and state one correct method, application or precaution.−

Karnaugh maps provide a visual method to simplify Boolean functions with a modest number of variables. Grouping adjacent 1s or 0s yields reduced SOP or POS expressions.

Answer framework: Map minterms/maxterms using Gray-code order, make largest valid power-of-two groups including edge adjacency, derive each term and verify coverage.

Important point: Do not group diagonal cells or groups of arbitrary size; only adjacency rules and powers of two apply.

Explain Arithmetic and comparison blocks and state one correct method, application or precaution.−

Half/full adders, subtractors and comparators combine logic functions to perform binary arithmetic and magnitude comparison. Multi-bit blocks cascade or use dedicated IC structures.

Answer framework: Begin with a one-bit truth table, derive sum/difference/carry/borrow outputs, then explain how the blocks extend to multiple bits.

Important point: Carry/borrow propagation affects multi-bit behaviour; test extreme and carry-in cases.

Explain MSI, multiplexing and programmable logic and state one correct method, application or precaution.−

Multiplexers select one input; decoders/encoders translate codes; ROM, PLA and PAL implement predefined or programmable logic. These blocks reduce discrete-gate complexity.

Answer framework: Identify data inputs, select inputs, enable and outputs; create a selection table before wiring or programming.

Important point: Enable polarity and unused inputs must be tied according to the device datasheet, not assumed.

Module 3 — Sequential Logic and State Systems

Explain Latches and flip-flops and state one correct method, application or precaution.−

Sequential circuits store state. Latches respond to enable level; flip-flops sample on a clock edge. SR, JK, D and T flip-flops differ in allowable input behaviour and use.

Answer framework: Use a truth/characteristic table, label clock/enable edge, trace present state to next state and note setup/hold requirements conceptually.

Important point: Asynchronous input changes near a clock edge can cause unreliable or metastable behaviour.

Explain Registers and counters and state one correct method, application or precaution.−

Registers store multi-bit words; shift registers move data between stages. Counters progress through a defined state sequence and may be ripple (asynchronous) or synchronous.

Answer framework: Draw state sequence and timing diagram, choose flip-flop type/count, identify reset and unused-state handling, then test the complete cycle.

Important point: Ripple counter propagation delays can create transient states that are unsuitable for some decoding applications.

Explain State-machine design and state one correct method, application or precaution.−

A state machine combines memory and combinational logic so outputs depend on state and possibly inputs. State diagrams/tables make behaviour explicit before implementation.

Answer framework: List states, inputs, outputs and reset state; create transition/output table, choose encoding and verify every input/state combination.

Important point: Do not omit invalid or reset/recovery conditions in a practical state-machine design.

Module 4 — Logic Families and Data Conversion

Explain Logic-family characteristics and state one correct method, application or precaution.−

Logic families differ in supply range, logic thresholds, speed, power, fan-out, noise margin and input/output structure. Device selection balances system requirements.

Answer framework: Compare candidate families using a specification table with voltage, delay, power, interfacing and environmental constraints.

Important point: Never mix logic families without checking threshold compatibility, supply requirements and output-drive capability.

Explain Digital-to-analog conversion and state one correct method, application or precaution.−

A DAC converts a digital code into an analog voltage/current. Weighted-resistor, R–2R and PWM approaches have different resolution, accuracy, speed and filtering needs.

Answer framework: State required range, resolution and update speed; calculate code step conceptually and choose a topology with an output-conditioning plan.

Important point: Resolution is not the same as accuracy; reference stability and nonlinearity also matter.

Explain Analog-to-digital conversion and state one correct method, application or precaution.−

An ADC samples an analog signal and converts it to a code. Flash, counter, SAR, sigma-delta and integrating types trade speed, resolution, noise performance and complexity.

Answer framework: Define input range, signal bandwidth, required resolution and conversion rate; select a type and state anti-aliasing/reference considerations.

Important point: Sampling without appropriate signal conditioning can create aliasing or invalid readings.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Binary Systems, Boolean Algebra and Gates.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Combinational Circuit Design.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Sequential Logic and State Systems.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Logic Families and Data Conversion.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Binary/octal/hexadecimal representation, signed numbers, Boolean laws, truth tables, standard forms and logic gates..
2. Develop a complete answer or practical plan for: SOP/POS, K-maps, adders/subtractors, comparators, encoders/decoders, multiplexers/demultiplexers, ROM/PLA/PAL..
3. Develop a complete answer or practical plan for: Latches, SR/JK/D/T flip-flops, triggering, excitation tables, registers, shift registers, ripple/synchronous counters and state machines..
4. Develop a complete answer or practical plan for: TTL/CMOS/ECL characteristics, fan-in/fan-out, noise margin, propagation delay; DAC/ADC types, specifications and selection..

LAST-MILE REVISION

Quick Revision

Module 1: Binary Systems, Boolean Algebra and Gates

Binary/octal/hexadecimal representation, signed numbers, Boolean laws, truth tables, standard forms and logic gates.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Combinational Circuit Design

SOP/POS, K-maps, adders/subtractors, comparators, encoders/decoders, multiplexers/demultiplexers, ROM/PLA/PAL.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: Sequential Logic and State Systems

Latches, SR/JK/D/T flip-flops, triggering, excitation tables, registers, shift registers, ripple/synchronous counters and state machines.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Logic Families and Data Conversion

TTL/CMOS/ECL characteristics, fan-in/fan-out, noise margin, propagation delay; DAC/ADC types, specifications and selection.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Number systems and codes	Digital systems represent information with discrete symbols, commonly binary bits.
Boolean algebra and truth tables	Boolean algebra models two-valued logic.
Logic gates and universal implementation	AND, OR, NOT, NAND, NOR, XOR and XNOR gates implement Boolean functions.
K-map minimisation	Karnaugh maps provide a visual method to simplify Boolean functions with a modest number of variables.
Arithmetic and comparison blocks	Half/full adders, subtractors and comparators combine logic functions to perform binary arithmetic and magnitude comparison.
MSI, multiplexing and programmable logic	Multiplexers select one input; decoders/encoders translate codes; ROM, PLA and PAL implement predefined or programmable logic.
Latches and flip-flops	Sequential circuits store state.
Registers and counters	Registers store multi-bit words; shift registers move data between stages.
State-machine design	A state machine combines memory and combinational logic so outputs depend on state and possibly inputs.
Logic-family characteristics	Logic families differ in supply range, logic thresholds, speed, power, fan-out, noise margin and input/output structure.
Digital-to-analog conversion	A DAC converts a digital code into an analog voltage/current.
Analog-to-digital conversion	An ADC samples an analog signal and converts it to a code.

LEARNING RESOURCES

References

Recommended digital-circuits references

1. M. Morris Mano and Michael D. Ciletti, Digital Design.
2. R. P. Jain, Modern Digital Electronics.
3. A. Anand Kumar, Fundamentals of Digital Circuits.

Approved public digital-electronics sources

- https://s3-ap-southeast-1.amazonaws.com/gtusitecirculars/Syallbus/BE04024041.pdf
- https://www.cl.cam.ac.uk/teaching/0708/DigElec/Digital_Electronics_pdf.pdf

The listed public sources support this corrected study handbook while official Course 3042 content is unavailable; they do not replace an official detailed syllabus.